Name: _____ **MTH234 Linear Algebra**
ID: _____ **1/2025 (Regular/International Section)**
22nd October 2025
3 hrs. (09:00 - 12:00 AM.)

MODULE 2 MOCK EXAM (Total 75 points + 45 Extra points)

Instruction: Students must provide all details of your work. You may use any class materials. Calculators and other materials are not allowed.

(1) (10 points) Let $V = P_3$ be the vector space of all polynomials of degree at most 3. Let W be the subset of V defined by

$$W = \{p(x) = ax^3 + bx^2 + cx + d \in P_3 \mid 2a - b + c - 2d = 0\}$$

Show that W is a subspace of P_3.

(2) (15 points) Let $S = \{v_1, v_2, v_3, v_4\}$ be a set of vectors in $\mathbb{R}^3$, where

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \; v_2 = \begin{bmatrix} -2 \\ -2 \\ 4 \end{bmatrix}, \; v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \; v_4 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$$

(a) Find a basis for the subspace $W = Span(S)$.

(b) Determine if the vector $u = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$ is in W. If it is, express u as a linear combination of the basis.

(3) (10 points) Find a basis for the solution space of the following homogeneous linear system:

$$\begin{aligned} x_1 + 2x_2 - x_3 + 3x_4 &= 0 \\ 2x_1 + 4x_2 - x_3 + 6x_4 &= 0 \\ -x_1 - 2x_2 + 3x_3 - 9x_4 &= 0 \end{aligned}$$

(4) (13 points) Let $B = \{b_1, b_2, b_3\}$ and $C = \{c_1, c_2, c_3\}$ be two bases for $\mathbb{R}^3$. The coordinate vectors of the basis vectors of C relative to the basis B are given as follows:

$$[c_1]_B = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \quad [c_2]_B = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}, \quad [c_3]_B = \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$$

(a) Find the change-of-coordinates matrix ${}_BP_C$ (from basis C to basis B).

(b) Find the change-of-coordinates matrix ${}_CP_B$ (from basis B to basis C).

(c) If a vector v has the coordinate vector $[v]_C = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$, find its coordinate vector $[v]_B$.

(5) (12 points) Consider the matrix A which depends on a parameter k:

$$A = \begin{bmatrix} 1 & 2 & k \\ 2 & 5 & 2k \\ 1 & 3 & 2 \end{bmatrix}$$

(a) Find the rank of A in terms of k.

(b) Find the value(s) of k for which the system $A\mathbf{x} = \mathbf{0}$ has non-trivial solutions.

(c) For the value(s) of k found in part (b), find a basis for the Null Space and a basis for the Column Space of A.

(6) (15 points) Let W be the subspace of $\mathbb{R}^4$ spanned by the vectors

$$w_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \quad w_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 2 \end{bmatrix}, \quad w_3 = \begin{bmatrix} 4 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

And let $v = \begin{bmatrix} -1 \\ 1 \\ 2 \\ 3 \end{bmatrix}$.

(a) Use the Gram-Schmidt process to find an orthonormal basis for W.

(b) Find the orthogonal projection of v onto W.

(c) Find the distance from the vector v to the subspace W.

(7) (Extra 15 points) Let $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \\ 5 \end{bmatrix} \right\}$ Consider $U = \operatorname{Span} S$ a subspace of $\mathbb{R}^4$.

(a) Find a basis for U.

(b) Find a basis for the orthogonal complement $U^{\perp}$.

(8) (Extra 15 points) Use the Gram–Schmidt process to produce an orthogonal basis and orthonormal basis for the column space of

$$A = \begin{bmatrix} -10 & 13 & 7 & -11 \\ 2 & 1 & -5 & 3 \\ -6 & 3 & 13 & -3 \\ 16 & -16 & -2 & 5 \\ 2 & 1 & -5 & -7 \end{bmatrix}.$$

(9) (Extra 15 points) Consider the following vectors in $\mathbb{R}^4$ with the standard inner product:

$$\mathbf{u} = \begin{bmatrix} -1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} 2 \\ 1 \\ -3 \\ 0 \end{bmatrix}, \qquad \mathbf{w} = \begin{bmatrix} a \\ -2 \\ 1 \\ 3 \end{bmatrix}, \qquad \mathbf{z} = \begin{bmatrix} 3 \\ -1 \\ 2 \\ a \end{bmatrix},$$

where $a \in \mathbb{R}$ is a parameter, and let $\mathbf{p} = \begin{bmatrix} 3 \\ 1 \\ 4 \\ -2 \end{bmatrix}$.

(a) For a general a, determine *all pairs* among $\{\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{z}\}$ that are orthogonal (if any). Also find the value(s) of a that make any additional pairs orthogonal.

(b) Let $a = 5$ and $W = \text{span}\{\mathbf{u}, \mathbf{z}\}$. Compute the orthogonal projection $\text{proj}_W(\mathbf{p})$ and the distance from $\mathbf{p}$ to W.

(c) Let $L = \text{span}\{\mathbf{v}\}$. Compute $\text{proj}_L(\mathbf{p})$.